

# EMU-ZEBU固件配置方法

## 1 概述

某些情况下，需要配置EMU仿真环境的固件，此处仅针对DRAMIF BANK HASH MASK的配置方法进行说明。

其余参数的配置，可直接参考固件发布说明中的信息进行操作。

## 2 配置方法

此处以firmware 0.7.2版本为基础进行整理。

- (1) firmware提供了DRAMIF BANK HASH MASK的配置方法：[SIPU FW V0.7.2固件发布说明](#)
- (2) 可以配置的DRAMIF BANK HASH MASK参数：

| IPOLY本原多项式取值 | dram bank interleave(Byte) | bankid_mask0                |        | bankid_mask1                |        | bankid_mask2                |        |
|--------------|----------------------------|-----------------------------|--------|-----------------------------|--------|-----------------------------|--------|
|              |                            | 计算方式                        | 寄存器取值  | 计算方式                        | 寄存器取值  | 计算方式                        | 寄存器取值  |
| 11           | 128                        | $((0x74e9) << 0) \& 0xffff$ | 0x74e9 | $((0x9d3a) << 0) \& 0xffff$ | 0x9d3a | $((0x3a74) << 0) \& 0xffff$ | 0x3a74 |
|              | 256                        | $((0x74e9) << 1) \& 0xffff$ | 0xe9d2 | $((0x9d3a) << 1) \& 0xffff$ | 0x3a74 | $((0x3a74) << 1) \& 0xffff$ | 0x74e8 |
|              | 512                        | $((0x74e9) << 2) \& 0xffff$ | 0xd3a4 | $((0x9d3a) << 2) \& 0xffff$ | 0x74e8 | $((0x3a74) << 2) \& 0xffff$ | 0xe9d0 |
|              | 1024                       | $((0x74e9) << 3) \& 0xffff$ | 0xa748 | $((0x9d3a) << 3) \& 0xffff$ | 0xe9d0 | $((0x3a74) << 3) \& 0xffff$ | 0xd3a0 |
| 13           | 128                        | $((0x5cb9) << 0) \& 0xffff$ | 0x5cb9 | $((0xb972) << 0) \& 0xffff$ | 0xb972 | $((0x2e5c) << 0) \& 0xffff$ | 0x2e5c |
|              | 256                        | $((0x5cb9) << 1) \& 0xffff$ | 0xb972 | $((0xb972) << 1) \& 0xffff$ | 0x72e4 | $((0x2e5c) << 1) \& 0xffff$ | 0x5cb8 |
|              | 512                        | $((0x5cb9) << 2) \& 0xffff$ | 0x72e4 | $((0xb972) << 2) \& 0xffff$ | 0xe5c8 | $((0x2e5c) << 2) \& 0xffff$ | 0xb970 |
|              | 1024                       | $((0x5cb9) << 3) \& 0xffff$ | 0xe5c8 | $((0xb972) << 3) \& 0xffff$ | 0xcb90 | $((0x2e5c) << 3) \& 0xffff$ | 0x72e0 |

将图中针对IPOLY=11的512B interleave参数，转换成firmware配置参数，如下：

代码块

```
1 # 按照ipoly11参数配置
2 0xe9d074e8d3a4
```

- (3) driver提供了查看DRAMIF BANK HASH MASK配置结果的方法：  
[https://gitlabsoft.siorigin.com/platform\\_sw/sio-drv/driver/test/-/blob/sipu-staging/utils/pcie\\_mem.c](https://gitlabsoft.siorigin.com/platform_sw/sio-drv/driver/test/-/blob/sipu-staging/utils/pcie_mem.c)

可自行编译，并移动至指定目录下。

(4) 可以查询DRAMIF BANK HASH MASK寄存器地址的链接:

[http://register.siorigin.com/data/sipu\\_v1p5/](http://register.siorigin.com/data/sipu_v1p5/)

以下是BANK0/BANK1/BANK2的HASH MASK寄存器地址:

代码块

```
1  BANK0_HASH_MASK
2  Absolute Address:0x4E_0020 Size:0x4
3  dramif bank0 hash mask
4
5  BANK1_HASH_MASK
6  Absolute Address:0x4E_0024 Size:0x4
7  dramif bank1 hash mask
8
9  BANK2_HASH_MASK
10 Absolute Address:0x4E_0028 Size:0x4
11 dramif bank2 hash mask
```

(5) 目前使用的firmware配置和系统内查询指令如下:

代码块

```
1  # firmware配置ROM参数
2  cd firmware_release/mscp/SIPU_EMU_A0_0_7_2/fw_creator_tool_src/tool/fw_creator/
3  python3.11 ./configuration.py -r ../../../../firmware_bin/SIPU_EMU_A0_0_7_2.rom -
4  M 0xe9d074e8d3a4
5  cd ../../../../../../../
6  source firmware_release/mscp/SIPU_EMU_A0_0_7_2/setup.csh zebu
7
8  # EMU/QEMU启动后, 进入QEMU
9  # .....
10 # 系统内查询配置结果, 0x4E0040用来查询interleave参数
11 sudo pci_mem -d 01:00.0 -b 0 -o 0x4E0020 -l 4
12 sudo pci_mem -d 01:00.0 -b 0 -o 0x4E0024 -l 4
13 sudo pci_mem -d 01:00.0 -b 0 -o 0x4E0028 -l 4
14 sudo pci_mem -d 01:00.0 -b 0 -o 0x4E0040 -l 4
15
16 # 配置结果查询输出, 与配置参数一致, 符合预期
17 ceshi@ceshi:~$ sudo pci_mem -d 01:00.0 -b 0 -o 0x4E0020 -l 4
18 /sys/bus/pci/devices/0000:01:00.0/resource0 opened.
19 Target offset is 0x4e0020, page size is 4096, display unit is 4 bytes
20 004e0020 0000d3a4
21 ceshi@ceshi:~$ sudo pci_mem -d 01:00.0 -b 0 -o 0x4E0024 -l 4
22 /sys/bus/pci/devices/0000:01:00.0/resource0 opened.
23 Target offset is 0x4e0024, page size is 4096, display unit is 4 bytes
```

```
24 004e0024 000074e8
25 ceshi@ceshi:~$ sudo pci_mem -d 01:00.0 -b 0 -o 0x4E0028 -l 4
26 /sys/bus/pci/devices/0000:01:00.0/resource0 opened.
27 Target offset is 0x4e0028, page size is 4096, display unit is 4 bytes
28 004e0028 0000e9d0
29 ceshi@ceshi:~$ sudo pci_mem -d 01:00.0 -b 0 -o 0x4E0040 -l 4
30 /sys/bus/pci/devices/0000:01:00.0/resource0 opened.
31 Target offset is 0x4e0040, page size is 4096, display unit is 4 bytes
32 004e0040 00000002
```